

Configuration of a Turing machine

Tape of M consists of a string $y \sqcup^\omega$ where $y \in \Gamma^*$.

A configuration is $\in Q \times \{y \sqcup^\omega \mid y \in \Gamma^*\} \times \mathbb{N}$
 (p, z, n)
 $Q \leftarrow$ state
 $n \geq 0$ current position of head on tape
current content of the tape.
 $\underbrace{\{y \sqcup^\omega \mid y \in \Gamma^*\}}_{\text{infinite string with a finite presentation.}}$

Start configuration - $(q, \vdash x \sqcup^\omega, 0)$
input string

Next configuration relation $\xrightarrow{\frac{1}{m}}$

For an infinite string z , let z_n be the n^{th} symbol in z

$S_b^n(z)$ - string obtained by substituting b for z_n .

Example $S_b^3(\vdash a b a a \dots) = \vdash a b b a \dots$

$$(p, z, n) \xrightarrow{\frac{1}{m}} \begin{cases} (q, S_b^n(z), n-1) & \text{if } \delta(p, z_n) = (q, b, L) \\ (q, S_b^n(z), n+1) & \text{if } \delta(p, z_n) = (q, b, R) \end{cases}$$

$\alpha \xrightarrow{*} \beta$: Reflexive transitive closure of $\xrightarrow{\frac{1}{m}}$

M accepts $x \in \Sigma^*$ if $(s, \vdash x \sqcup^w, 0) \xrightarrow{*}_M (t, y, n)$

$$L(M) = \{x \in \Sigma^* \mid M \text{ accepts } x\}$$

M rejects $x \in \Sigma^*$ if $(s, \vdash x \sqcup^w, 0) \xrightarrow{*}_M (r, y, n)$

M **halts** on input $x \in \Sigma^*$ if it either accepts x or rejects x .

M **loops** on input $x \in \Sigma^*$ if M neither accepts nor rejects x .

Total TM: A Turing machine that halts on all inputs is called a total TM.

$A \subseteq \Sigma^*$ is **recursive** if $A = L(M)$ for a total TM M .

$A \subseteq \Sigma^*$ is **recursively enumerable (r.e.)** if $A = L(M)$ for some TM M (need not be a total TM).

Ex. $A = \{a^n b^n c^n \mid n \geq 0\}$ is recursive.

Example. $A = \{ww \mid w \in \{a,b\}^*\}$ — not a CFL

A is recursive since we can define a total TM M s.t. $L(M) = A$.

Working of M on input x .

- Scan (left to right) till first $\sqcup$ - symbol
replace $\sqcup$ with $|$ and check that number of symbols in x is even
- In each pass from right to left, M marks the first unmarked a, b with $\acute{a}, \acute{b}$
- In each pass from left to right, it marks the first unmarked a, b with $\grave{a}, \grave{b}$

Ex. Tape Content $\Gamma = \{a, b, \vdash, \sqcup, \dashv, \grave{a}, \grave{b}, \acute{a}, \acute{b}\}$

$\vdash a a b b b a a b b b \sqcup \sqcup \sqcup \sqcup$

$\vdash a a b b b a a b b b \dashv \sqcup \sqcup \sqcup$

$\vdash \grave{a} \grave{a} \grave{b} \grave{b} \grave{b} \acute{a} \acute{a} \acute{b} \acute{b} \acute{b} \dashv \sqcup \sqcup \sqcup$

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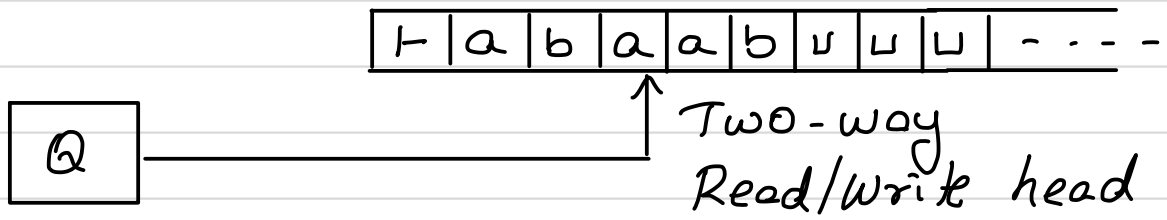
| a a b b b a a b b b $\sqcup$ $\sqcup$ $\sqcup$ $\sqcup$

| a a b b b a a b b b | $\sqcup$ $\sqcup$ $\sqcup$ $\sqcup$

| $\grave{a}$ $\grave{a}$ $\grave{b}$ $\grave{b}$ $\grave{b}$ $\acute{a}$ $\acute{a}$ $\acute{b}$ $\acute{b}$ $\acute{b}$ | $\sqcup$ $\sqcup$ $\sqcup$ $\sqcup$

| $\sqcup$ $\sqcup$ $\grave{b}$ $\grave{b}$ $\grave{b}$ $\sqcup$ $\sqcup$ $\acute{b}$ $\acute{b}$ $\acute{b}$ | $\sqcup$ $\sqcup$ $\sqcup$ $\sqcup$

Turing machines - equivalent models.



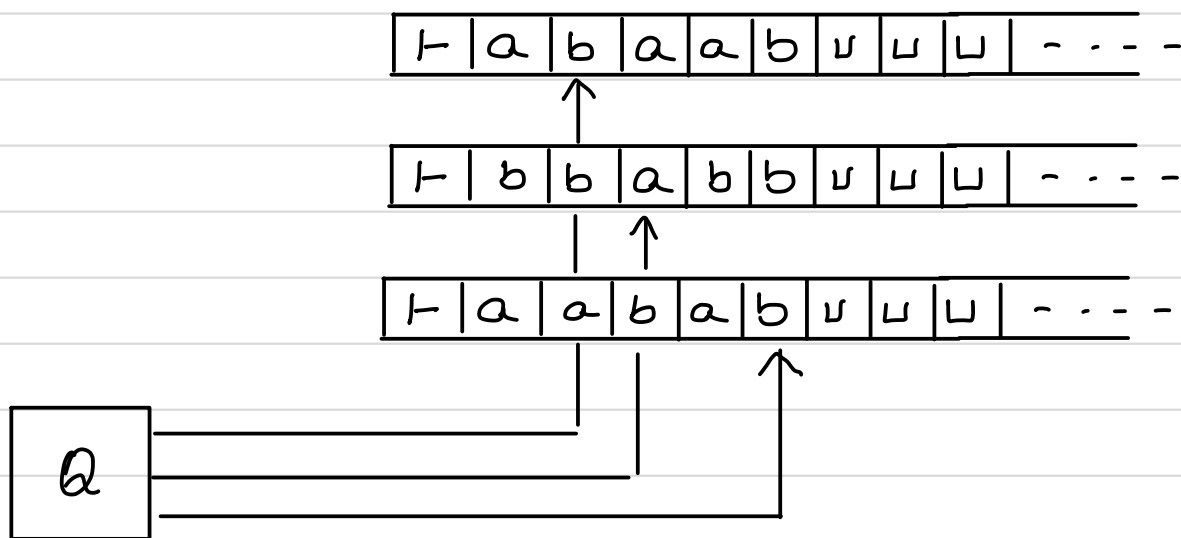
Tape with multiple tracks.

	␣	a	a	b	a	b	␣	␣	
␣	␣	b	b	b	a	b	b	␣	
	␣	b	a	a	a	b	a	␣	

Tape symbol is a triple $(c, d, e) \mapsto$

c
d
e

Multiple tapes



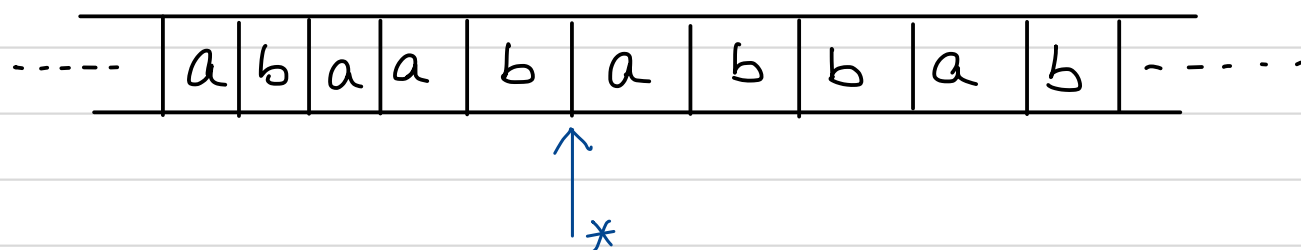
$$\delta: Q \times \Gamma^3 \rightarrow Q \times \Gamma^3 \times \{L, R\}^3$$

	␣	a	b	a	a	b	␣	␣	
␣	␣	b	b	a	b	b	␣	␣	
	␣	a	a	b	a	b	␣	␣	

Consider the tape alphabet

$$\Sigma \cup \{\sqcup\} \cup (\Gamma \cup \Gamma')^3 \quad \text{where } \Gamma' = \{\hat{a} \mid a \in \Gamma\}$$

Two way infinite tape



⌈	b	a	a	b	a	---
	a	b	b	a	b	---

Simulate top track when head is on the left of * and simulate bottom track when head is on the right of *

Power of the model - Universal Turing machine.

Fix an encoding scheme of TM over some alphabet
Say $\{0,1\}$

Any encoding is fine as long as it is possible for another TM to take as input the encoded string and decode the description.

An example encoding scheme

$0^n | 0^m | 0^k | 0^s | 0^t | 0^r | 0^u | 0^v |$

M has n states $\downarrow$ m tape symbols of which first k are input symbols

Similar encoding possible for transitions.

Important properties of the encoding scheme

- Able to encode all TMs upto isomorphism.
- Easy to interpret.

Universal Turing Machine

$$L(U) = \{ \overline{M\#x} \mid x \in L(M) \}.$$

encoding of M



encoding of input string x
 x is over the input alphabet
of M .

Symbol in U 's input alphabet
to delimit M and x .